Problem 3 $\Leftarrow$ Proposition from Lecture 10 (in particular (c))

Fact 1 $\alpha \in k$: finite field of char. p

$|k| = p^d$

$Fr^m(\alpha) = \alpha \iff Fr^{\gcd(m,d)}(\alpha) = \alpha$

Fact 2 (Problem 2) : $|k| = p^d$, $f(x) \in \mathbb{F}_p[x]$ irred. of deg r

TFAE: 1 $r \mid d$

2 $f(x)$ has r distinct zeros in k

3 $f(x)$ has a zero in k.

$\cdot\ \alpha \in k$: finite of char p

Def$^{\underline{n}}$ $m_\alpha(x)$ is $\boxed{\text{the}}$ irreducible polynomial with leading coeff 1 s.t.

$$(m_\alpha(x)) = \ker ev_\alpha$$

$$\begin{tikzcd}
\mathbb{F}_p[x] \arrow[d] \arrow[drr, two heads, "ev_\alpha"] & & \\
\mathbb{F}_p[x] / (m_\alpha(x)) \arrow[rr] & & k
\end{tikzcd}$$

Fact 3 : TFAE for $f(x) \in \mathbb{F}_p[x]$

1. $f(\alpha) = 0 \in k$
2. $f(x) \in \ker ev_\alpha$
3. $m_\alpha(x) \mid f(x)$

Linear algebra

k: field

Def^{n} A k-vector space

or a vector space over k

is a tuple $(V, +, 0, \cdot)$

where: $\boxed{(a)\ (V, +, 0) \text{ is an additive abelian group}}$

$\begin{pmatrix}\text{scalar}\\ \text{multiplication}\end{pmatrix} \to$ (b) $\cdot : k \times V \longrightarrow V$

$(a, v) \longmapsto a \cdot v$

s.t. (1) (Distributivity)

$a \in k,\ v_1, v_2 \in V$

$a \cdot (v_1 + v_2) = a \cdot v_1 + a \cdot v_2$

(2) (Associativity)

$a, b \in k,\ v \in V$

$a\cdot(b\cdot v) = (ab)\cdot v$

(3) (Identity for scalar multiplication)

$\forall v \in V$

$1\cdot v = v$

Rmk: $a \in k^{\times}$

$a^{-1}\cdot(a\cdot v)$

$= (a^{-1}a)\cdot v$

$= 1\cdot v$

$= v$

Examples

(1) $\{0\}$: k-vector space with

$a\cdot 0 = 0, \ \forall a \in k.$

(2) k :

$$\begin{aligned} k \times k &\longrightarrow k \\ (a, b) &\longmapsto a\cdot b \end{aligned}$$

(3) V, W : k-vector spaces

$V \times W$: direct product

is also a k-vector space with

co-ordinatewise addition &
scalar multiplication

(4) $\underbrace{k \times \cdots \times k}_{n\text{-times}} \stackrel{\text{defn}}{=} k^n$

$k^n = \{(a_1, a_2, \ldots, a_n) : a_i \in k\}$

(5) $k[x]$: k-vector space

$$k \times k[x] \longrightarrow k[x]$$
$$(a, f(x)) \longmapsto a \cdot f(x)$$

(6) $k \leq R \leftarrow$ commutative ring

sub-ring

then R is a k-vector space

$$k \times R \longrightarrow R$$
$$(a, r) \longmapsto a \cdot r$$

(7) In the situation of (6)

$I \leq R$: ideal

Then

I is a k-subspace of R

(8) In <u>particular</u>, $k \leq L$: field ext.

then L is a k-vector space.

<u>Def^n</u> V: k-vector space

$W \leq V$: subgroup

W is a **k-subspace** (or just **subspace**)

if $\forall a \in k, w \in W$,

$a \cdot w \in W$

(i.e. closed under $\cdot$)

<u>Example:</u> (1) $\{0\} \leq V$ subspace (2) $V \leq V$ subspace

Rmk: $k \leq L$: field ext.

Then any L-vector space also has the structure of a k-vector space.

Defn: V: k-vector space

$X \subseteq V$: subset

The k-subspace generated by X

$$\langle X \rangle_k \leq V$$

is the smallest subspace of V containing X

i.e. If $W \leq V$ is any subspace containing X, then

$$\langle X \rangle \leq W.$$

Rmk: (1) $W_i \leq V$ are k-subspaces $i \in I$ (I is an indexing set)

Then

$$\bigcap_{i \in I} W_i \leq V$$

is also a k-subspace

(2) $I_X = \left\{ \begin{array}{l} \text{subspaces } W \leq V \\ \text{s.t. } X \subseteq W \end{array} \right\}$

$(X \subseteq V)$

$$\ni$$

$$V$$

(top-down)

$$\bigcap_{W \in I_X} W = \langle X \rangle.$$

Defn $v_1, v_2, \ldots, v_n \in V$

A **k-linear combination** of $v_1, \ldots, v_n$ is an element of the form $v = a_1 v_1 + \ldots + a_n v_n \in V$ with $a_1, a_2, \ldots, a_n \in k$

(3) (Bottom-up)

$$\langle X \rangle = \left\{ \sum_{i=1}^{n} a_i v_i : \begin{array}{l} a_i \in k \\ v_i \in X \end{array} \right\}$$

$= \{$ k-linear combinations of elements of X $\}$

Fact V: k-v.s.

(a) $v \in V$

$0 \cdot v = 0_V \in V$

(b) $v \in V$

$(-1) \cdot v = -v \in V$

<u>Pf:</u> Enough to show:

(a) RHS is a $\mathbb{k}$-subspace of V

(b) RHS is <u>contained</u> in every subspace containing X.